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Sky Care Solutions

Web App/Mobile App- Technical Doc

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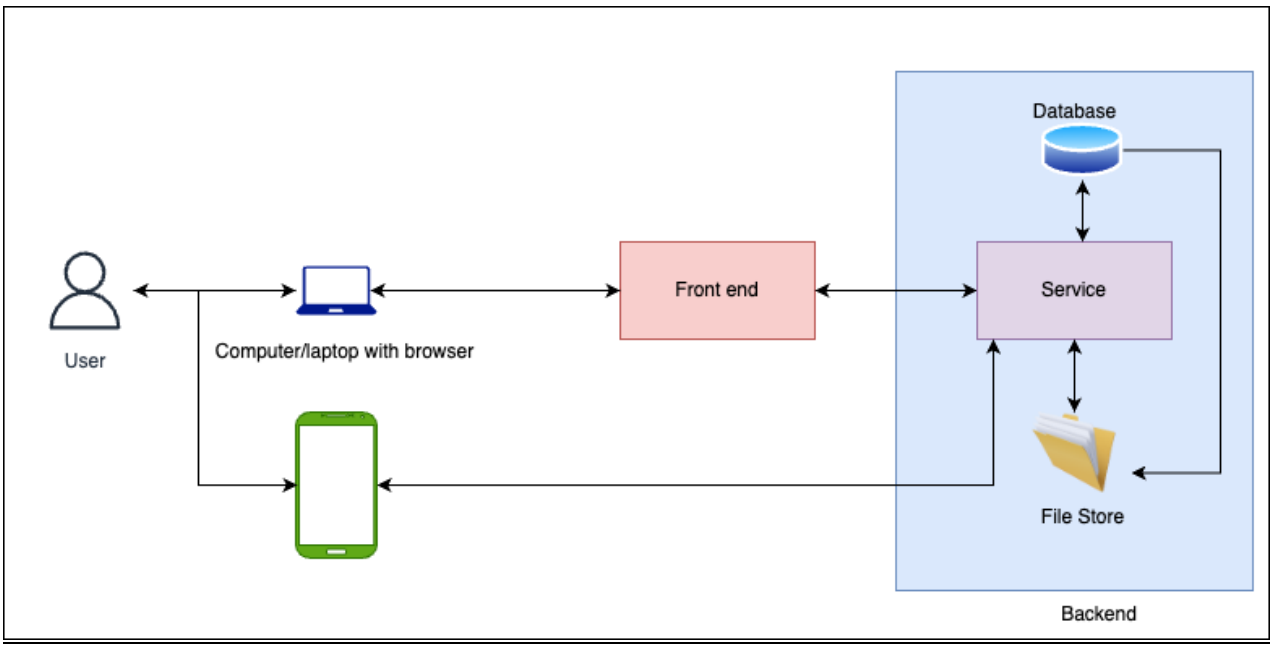
Table of Contents

1.0 System Architecture....2
1.1 Sky Care Solutions web app architectural diagram....2
2.0 Requirement Analysis....2
2.1 Functional Requirements of Sky Care Solutions Web App....2
2.4 Non-Functional Requirements....4
2.5 Non-Functional Requirements of Sky Care Solutions Web App....4
3.0 User Stories....5
3.3 User stories of Sky care solutions....5
4.0 Deployment Diagram....7
4.1 Overview....8
4.2. Hosting Infrastructure....8
4.3.Pipeline Steps:....8
4.4. Runtime Deployment on OVH VPS....8
Inside the VPS:....9
4.5. Environment Architecture....9
4.6. Nginx Reverse Proxy....10
4.7. Security Measures....10
5.0 Mobile Application....10
5.1. Overview....10
5.2. Mobile App Details....10
5.3. Development Stack....10
5.4. Platform-Specific Hosting & Deployment....11
5.5. Integration with Backend....11
5.6. Security & Compliance....11
5.7. Monitoring & Updates....12

1.0 System Architecture

An architectural diagram is a visual representation of how different parts of a software system are organized and interact with each other. It shows the structure of the web application, including how data flows between different components like the front end (user interface), back end (server), database, and third-party services.

1.1 Sky Care Solutions web app architectural diagram.



2.0 Requirement Analysis

Requirements in application development refer to stakeholders' documented needs and expectations (clients, users, and developers) for a software project. These requirements describe the software's behavior, functionality, and constraints and form the foundation for the design, development, and testing phases. Requirements are categorized into functional and non-functional requirements.

2.1 Functional Requirements of Sky Care Solutions Web App

These define the specific behavior and features the software must support. They outline what the system should do to fulfill the user and business needs.

Authentication and Authorization

- The system must allow Super Admin, Admin, Staff, and Support Workers to log in using email and password (US001, US007, US023, US031).
- Super Admins can create, view, edit, and delete Admin accounts, but cannot delete or edit other Super Admins (US002, US003).

- Support workers should be able to self-register, and Admins must review and approve or reject the registration (US028, US029, US021).

Profile Management

- All users (Super Admins, Admins, Staff, Support Workers) must be able to view and update their profile details (US004, US008, US024, US032).
- All users should be able to change their passwords (US005, US009, US025, US033).

Client Management

- Admins must be able to add, edit, and delete client details (US010, US011).
- Admins can create, edit, and update client service schedules while keeping previous records visible (US012, US013).
- Admins can assign support workers to scheduled client services (US014).
- Admins should be able to broadcast service schedules to support workers (US015).

Staff Management

- Admins must be able to add, edit, and delete staff details, including police certificates, qualifications, working with children's checks, and other compliance documents (US016, US017, US018, US019).
- Admins can assign shifts to staff and notify them (US020).
- Admins must be able to view expired police certificates for staff and take necessary actions (US022).

Shift Management

- Staff should be able to view their assigned shifts, submit timesheets with shift notes for payroll, and track their shifts (US026, US027).
- Support workers should be able to apply for available shifts and choose based on their availability (US034, US035).
- Support workers can view their previous tasks and shift history (US030, US038).

Document and Compliance Management

- Support workers can upload compliance documents like police certificates, NDIS screening, and others to complete their profiles (US036).
- Admins can mark compliance documents as compliant or non-compliant, and support workers can view the status (US039).
- Support workers should receive email notifications for available shifts (US037).

Dashboard and Notifications

- All users should have access to a dashboard that displays relevant information such as schedules, compliance status, tasks, and more (US040).

- Email notifications must be sent to support workers for available shifts and important updates (US037).

2.4 Non-Functional Requirements

These specify how the system should behave, focusing on the software's performance, security, usability, and other operational aspects. Non-functional requirements typically address how the system will meet the functional requirements.

2.5 Non-Functional Requirements of Sky Care Solutions Web App

- **Performance**
  The system should be able to handle multiple users (Super Admins, Admins, Staff, and Support Workers) logging in simultaneously without performance degradation.
  Shift and schedule updates should be reflected in real time to prevent any delays for support workers receiving notifications.

- **Usability**

  The user interface must be intuitive and user-friendly to accommodate users with varying levels of technical skills.
  The dashboard for all user roles must display clear and concise information, ensuring users can quickly access their tasks, schedules, and other relevant data.
- **Security**
  Passwords must be stored securely using industry-standard encryption methods.
  The system must enforce role-based access control, ensuring users can only access the data and functionalities relevant to their role.
  Sensitive data like compliance documents and personal information should be encrypted in storage and during transmission.
- **Scalability**
  The system should be designed to scale easily as more support workers, staff, and clients are added without affecting performance.
  The system should support an increasing number of shifts, schedules, and document uploads as the organization grows.
- **Reliability and Availability**
  The system should maintain a high availability rate to ensure users can access it at all times.
  Regular backups must be implemented to prevent data loss in case of failures.
- **Maintainability**
  The system must be built in a modular way, allowing for future updates, including adding new functionalities, fixing bugs, and improving the user experience without significant downtime.
  Proper documentation must be maintained for all modules to ensure easy troubleshooting and updates.
- **Compliance and Legal**

The system must comply with relevant data protection regulations such as GDPR and Australian Privacy Law, given that sensitive personal and compliance data will be stored. Regular audits must be conducted to ensure compliance with organizational policies and legal requirements.

3.0 User Stories.

User stories are short, simple descriptions of a feature or functionality written from the perspective of the end-user or customer. They focus on who the user is, what they need, and why they need it, making them more human-centered.

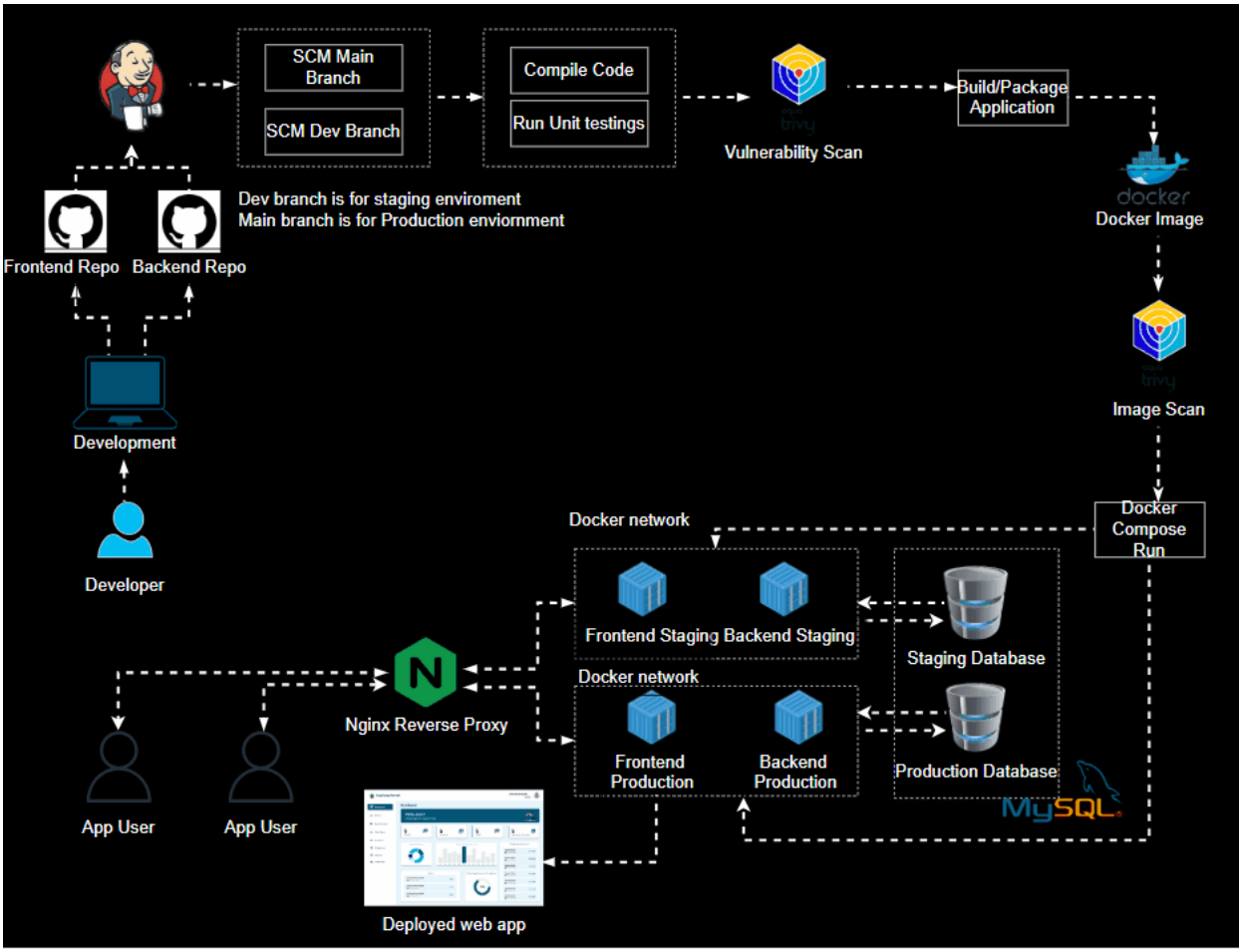
3.3 User stories of Sky care solutions.

User Story ID	User Type	User Story
US001	Admin	As an admin, I can be able to login to the system by using email and the password.
US002	Admin	As an admin, I can be able to add new admins to the system by entering new admin details
US003	Admin	As an admin, I can be able to view, edit, and delete all other admins (Cannot delete or edit super admins).
US004	Admin	As an admin, I can change my account details (Profile).
US005	Super Admin	As an admin, I can change my password
US006	Admin	As an admin, I have all the privilege of doing admin tasks
US007	Admin	As an admin, I can be able to login to the system by using email and the password.
US008	Admin	As an admin, I can change my account details (Profile).
US009	Admin	As an admin, I can change my password
US010	Admin	As an admin, I want to add new client details so that I can manage client records effectively.
US011	Admin	As an admin, I can edit and delete client.
US012	Admin	As an admin, I want to add a service schedule for a client so that I can organize services based on client needs.
US013	Admin	As an admin, I want to edit a client's service schedule so that I can update the schedule while keeping previous records.
US014	Admin	As an admin, I want to assign a support worker to a scheduled client service so that the client's needs are met.
US015	Admin	As an admin, after creating a schedule, I can broadcast it to all relevant details to support workers.
US016	Admin	As an admin, I want to add staff details so that I can manage employee information in one place.
US017	Admin	As an admin, I want to upload police certificates for staff so that I can keep track of compliance documents.
US018	Admin	As an admin, I want to upload qualifications and certificates for staff so

		that I can verify their skills and credentials.
US019	Admin	As an admin, I want to add a staff member working with children's check details so that I can track compliance.
US020	Admin	As an admin, I want to assign shifts to staff so that I can ensure staffing needs are met.
US021	Admin	As an admin, I can review the support worker profile and approve or reject him from the system.
US022	Admin	As an admin, I want to get the expired police certificates for staff members.
US023	Admin	As an Admin, I want to create care plans, edit them also
US024	Admin	As an Admin, I want to I can be able to upload documents and forms
US025	Admin	As an Admin, I want to add an hourly rate for the caregiver.
US026	Admin	As an Admin, I want to view the payment details (total Payments) of a caregiver
US027	Admin	As an Admin, I want to generate Employee time sheet reports.
US028	Admin	As an Admin, I want to add new roles and manage roles within the system
US029	Support Worker	As a Support worker, I want to be able to upload and download documents
US030	Support Worker	As a support worker, I want to upload consent according to the tasks/caretakers
US031	Support Worker	As a staff, I can be able to login to the system by using email and the password.
US032	Support Worker	As a staff, I can change my account details (Profile).
US033	Support Worker	As a staff, I can change my password
US034	Support Worker	As a staff member, I want to view my assigned shifts so that I know when and where I need to work.
US035	Support Worker	As a staff member, I want to submit my timesheet including shift notes so that my work hours are recorded for payroll.
US036	Support Worker	As a support worker, I can self-register to the system.
US037	Support Worker	To work with the company I have to wait until the admin approves my registration
US038	Support Worker	I can view my previously assigned tasks
US039	Support Worker	As a Support Worker, I can be able to login to the system by using email and a password.
US040	Support Worker	As a Support Worker, I can change my account details (Profile).
US041	Support Worker	As a Support Worker, I can change my password
US042	Support Worker	As a support worker, I can apply for available shifts

US043	Support Worker	As a support worker, I want to view available shifts so that I can choose the ones that fit my schedule.
US044	Support Worker	As a support worker, I want to upload compliance documents (e.g., police certificates, NDIS screening) so that I can ensure my profile is complete.
US045	Support Worker	As a support worker, I want to receive email notifications about available shifts so that I can respond in a timely manner.
US046	Support Worker	As a support worker, I want to see my shift history on the dashboard so that I can track my work.
US047	Support Worker	As a support worker, I want to see the status of my compliance documents so that I know if I meet all requirements.
US048	All	All the users have an interactive dashboard to view information

4.0 Deployment Diagram



4.1 Overview

Skycare is a web-based application deployed using Docker-based infrastructure. It follows a CI/CD model to ensure consistent and secure delivery of application updates.

4.2. Hosting Infrastructure

- **Hosting Provider**: OVH Cloud
- **Hosting Type**: Linux-based VPS (Virtual Private Server)
- **VPS Configuration**:
  - 4 vCPUs
  - 16 GB RAM
  - Ubuntu 22.04 LTS
- **Public IP**: Configured with domain name (e.g., skycare.au)
- **Services Deployed Inside VPS**:
  - Jenkins (for CI/CD)
  - Docker & Docker Compose
  - Nginx (as reverse proxy)
  - MySQL (for staging & production databases)
  - Frontend & Backend containers (for both staging and production)

4.3.Pipeline Steps:

1. **Build**: Compile frontend/backend code.
2. **Unit Testing**: Runs automated test suites.
3. **Trivy Static Scan**: Checks codebase for security flaws.
4. **Docker Image Build**: Frontend and backend are containerized.
5. **Trivy Image Scan**: Analyzes Docker images for vulnerabilities.
6. **Docker Compose Deployment**: Deploys containers to correct environment.

4.4. Runtime Deployment on OVH VPS

The OVH VPS is structured with the following services and layers:

Inside the VPS:

```
/opt/
│
├── jenkins/              --> Jenkins (CI/CD server)
├── docker/               --> Docker & Docker Compose setup
│   ├── staging/
│   │   ├── docker-compose.yml   --> Staging services
│   └── production/
│       ├── docker-compose.yml   --> Production services
├── nginx/
│   ├── conf.d/
│   │   ├── staging.conf
│   │   └── production.conf
│   └── ssl/              --> TLS/SSL certificates
├── databases/
│   ├── staging-mysql/
│   └── production-mysql/
```

4.5. Environment Architecture

Staging Environment:

- **Containers**:
  - frontend-staging
  - backend-staging
- **Database**:
  - MySQL instance for staging only
- **Purpose**:
  - Feature testing, QA validation
- **Access**:
  - Internally exposed or accessible via staging subdomain

Production Environment:

- **Containers**:
  - frontend-production
  - backend-production
- **Database**:
  - Separate MySQL instance for production data
- **Purpose**:
  - Live system serving end users
- **Access**:
  - Publicly available via production domain

4.6. Nginx Reverse Proxy

- Installed on the OVH VPS and configured to:
  - Route traffic based on domain/subdomain or path
  - Terminate SSL via Let's Encrypt
  - Proxy requests to internal Docker networks

4.7. Security Measures

- **Trivy Static Scan**: Scans code before building.
- **Trivy Image Scan**: Checks Docker image vulnerabilities.
- **Environment Isolation**: Staging and production run in different networks and databases.
- **Firewall & SSH Access**:
  - Only port 80/443 exposed
  - SSH access restricted via key authentication

5.0 Mobile Application

5.1. Overview

SkyCare also includes a **cross-platform mobile application** designed to provide a seamless experience for both Android and iOS users. The mobile app interacts with the same backend services as the web application, enabling users to manage their healthcare-related data, access services, and receive real-time updates.

5.2. Mobile App Details

Attribute	Description
App Name	**SkyCare**
Platform	Android & iOS
Development Framework	**React Native** with **Expo**
Backend Integration	Connects to the production backend APIs deployed on OVH VPS
Data Format	JSON via RESTful API
Auth Mechanism	Token-based authentication (e.g., JWT)

5.3. Development Stack

- **React Native**: Used for building cross-platform UI components.
- **Expo**: Provides development tools, OTA updates, push notifications, and simplified native builds.
- **Backend Communication**:
  - The app uses axios for API calls to the production backend hosted on the OVH VPS.
  - Configured with environment-specific API endpoints (e.g., staging vs production).

5.4. Platform-Specific Hosting & Deployment

Android Deployment

Item	Value
Store	**Google Play Store**
App Type	Public
Play Store Link	https://play.google.com/store/apps/details?id=com.vishath.skycare_mobile_app&pcampaignid=web_share
Build Process	Via eas build --platform android (Expo EAS Build)
Signing	Keystore managed via EAS or Play Console
Release Strategy	Versioned releases, staged rollout enabled

iOS Deployment

Item	Value
Store	**Apple App Store (Unlisted App)**
App Type	**Unlisted (Not publicly searchable)**
App Store Link	*https://apps.apple.com/us/app/skycare/id6742375251*
Build Process	Via eas build --platform ios (Expo EAS Build)
Provisioning	Managed with Apple Developer Account
Distribution	Only accessible via direct app link

Note: An unlisted app on iOS is distributed via a direct link but is not visible in public App Store search results. It is ideal for controlled access (e.g., internal teams, partners, or specific user groups).

5.5. Integration with Backend

- The mobile app communicates with the same production backend services as the web app.
- Features include:
  - User login and authentication
  - Profile and data management
  - Real-time health updates or notifications (Expo push notifications)
  - Appointment scheduling and medical records (if supported)
- All API requests go through HTTPS using the Nginx reverse proxy on the OVH VPS.

5.6. Security & Compliance

- All API traffic is encrypted via TLS/SSL.
- Authentication is done using JWT or secure token mechanisms.
- Expo SecureStore is used for securely storing sensitive data like tokens on the device.
- App permissions are minimal and user-consent driven (e.g., push notifications).

5.7. Monitoring & Updates

- **Over-the-Air (OTA) Updates**:
  - Enabled via Expo's OTA service for non-breaking updates (JS-only changes).
  - Allows instant delivery of bug fixes and UI changes without app store resubmission.